



# Commitment to Singlehood: Testing Measurement Invariance across Groups

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## COMMITMENT TO SINGLEHOOD

- Defined as one's intent to stay single, including a psychological attachment and a long orientation toward singlehood
- Evidence that the Investment Model for close relationships can be extended to understand commitment to singlehood and its predictors—singlehood satisfaction, alternatives, and investment to singlehood (Beauparlant et al., 2023; Rusbult, 1980)
- However, singlehood experiences can vary by gender, age, political orientation, relationship history (Apostolou et al, 2020; Fangmeier et al., 2020; Oh et al., 2021; Park et al., 2022)
- To ensure any observed group differences in commitment and its predictors are meaningful (vs. reflecting measurement bias), we tested for **measurement invariance** (Luong & Flake, 2021)

## METHOD

### Participants

- $N = 705$  single participants recruited through Prolific

### Measures

- Investment Model Scale for Singles (IMS-S; Beauparlant et al., 2024)
  - Table 1 shows items and corresponding subscales
    - singlehood satisfaction (S)
    - alternatives to singlehood (A)
    - investment (I)
    - commitment (C)

## Table 1. IMS-S Items

- I feel satisfied with being single. (S)
- Being single is much better than others' relationships. (S)
- Being single is close to ideal. (S)
- The people I might become involved with are very appealing. (A)
- Dating is appealing to me. (A)
- For me, being in a relationship would be an attractive alternative to being single. (A) †
- I have put a great deal into my single lifestyle that I would lose if I were to start a relationship. (I)
- Many aspects of my life have become linked to being single (recreational activities, etc.), and I would lose all of this if I were to start a relationship with someone. (I)
- My relationship with friends and family would be complicated if I were to start dating someone (e.g., being single is an important part of my friendships). (I)
- I am committed to being single. (C)
- I feel very attached to being single -- very strongly linked to my single lifestyle. (C)
- I am oriented towards the long-term future of being single (for example, I imagine being single several years from now). (C)
- I intend to stay single. (C)

Note. CFA indicated poor model fit for the initial four-factor structure. Follow-up analyses suggested modeling the structural relationship between predictors and commitment or modeling predictors (S, A, I) and commitment (C) separately. † flagged for removal.

Table 2. Measurement Invariance across Groups

| Groups                            | Commitment                             |      |        |        |
|-----------------------------------|----------------------------------------|------|--------|--------|
|                                   | Configural                             | Weak | Strong | Strict |
| Gender <sup>a</sup>               | ✓                                      | ▲    | ✓      | ✓      |
| Age <sup>b</sup>                  | ✓                                      | ✓    | ▲      | ▲      |
| Relationship History <sup>c</sup> | ✓                                      | ✓    | ✓      | ✓      |
| Political <sup>d</sup>            | ✓                                      | ✓    | ✓      | ✓      |
|                                   | Satisfaction, Alternatives, Investment |      |        |        |
|                                   | Configural                             | Weak | Strong | Strict |
| Gender                            | ✓                                      | ✓    | X      | ▲      |
| Age                               | ✓                                      | ✓    | ▲      | ✓      |
| Relationship History              | ✓                                      | ✓    | ✓      | ✓      |
| Political                         | ✓                                      | ✓    | ✓      | ✓      |

**Note.** ✓ = the more constrained model fit significantly worse. ▲ = 1/3 index exceeded the cutoff ( $\Delta RMSEA$  or  $\Delta SRMR$ ). X = multiple model fit indices exceeded cutoffs, i.e., the more constrained model fit significantly worse.

<sup>a</sup>**Gender:** Male (i.e.,  $n = 351$ ) vs. Female ( $n = 339$ )

<sup>b</sup>**Age:** Young adults (i.e., ages 18-39;  $n = 426$ ) vs. Middle+ adults (i.e., ages 40-77;  $n = 278$ )

<sup>c</sup>**Number of previous relationships:** Few (i.e., 1-3;  $n = 177$ ) vs. Many (i.e., 4+;  $n = 156$ )

<sup>d</sup>**Political orientation:** Liberal ( $n = 487$ ) vs. Conservative ( $n = 218$ )

Figure 1. Group difference in commitment and its predictors

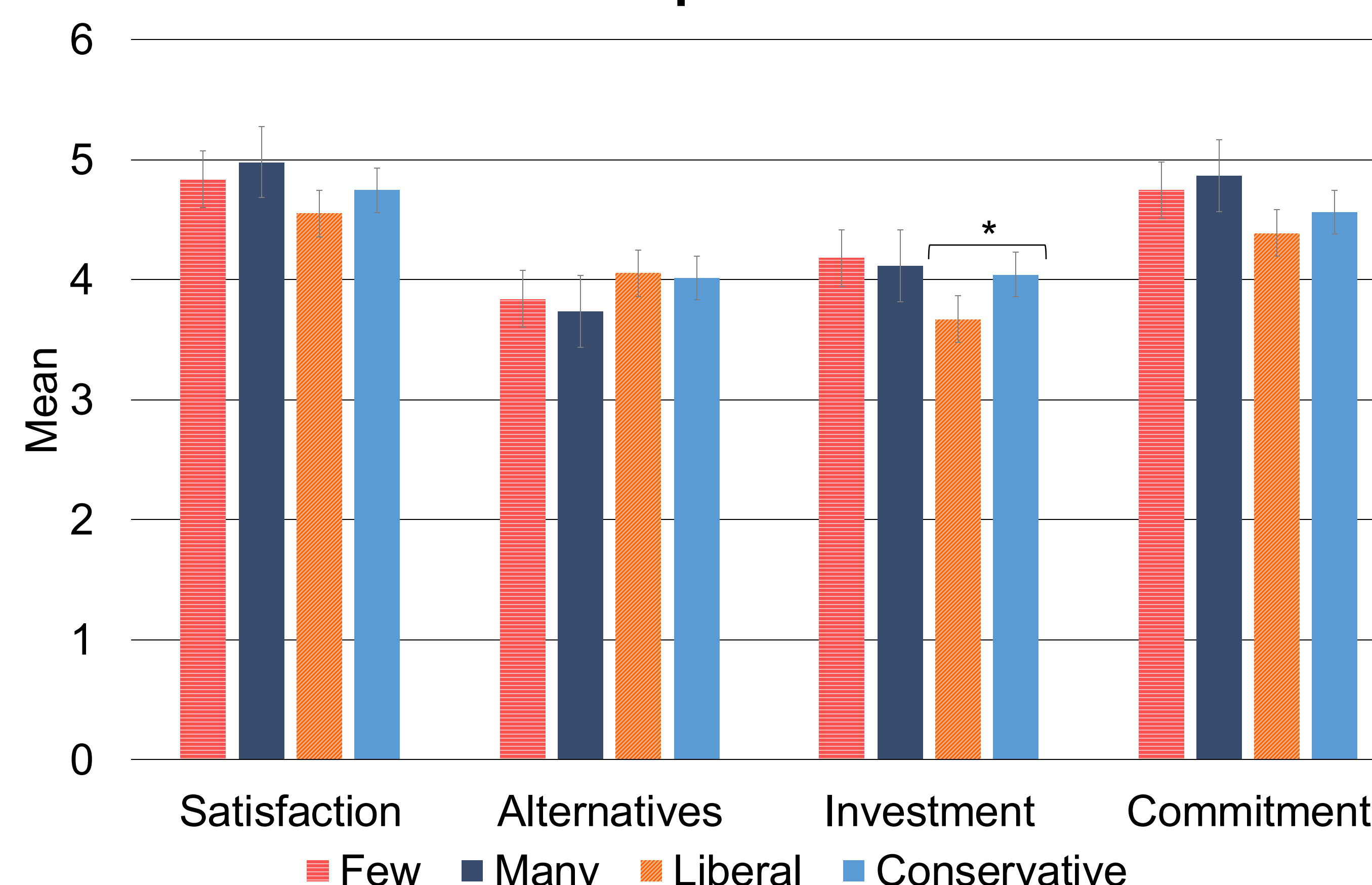


Table 3. Group Comparisons

| Groups               | $t$           | $df$       | $p$                 | $t$           | $df$       | $p$          |
|----------------------|---------------|------------|---------------------|---------------|------------|--------------|
|                      |               |            | <i>Satisfaction</i> |               |            |              |
| Relationship History | -0.970        | 327        | 0.330               | 0.357         | 319.17     | 0.720        |
| Political            | <b>-1.609</b> | <b>703</b> | <b>0.110</b>        | <b>-2.999</b> | <b>703</b> | <b>0.004</b> |
|                      |               |            | <i>Alternatives</i> |               |            |              |
| Relationship History | 0.646         | 326.96     | 0.530               | -0.707        | 322.47     | 0.480        |
| Political            | 0.344         | 703        | 0.730               | -1.306        | 703        | 0.192        |

## Plan for Invariance Testing

- Configural (same factor structure)
- Weak (equal factor loadings)
- Strong (equal intercepts)
- Strict (equal residuals)

## Decision Criteria

- CFI > .90, RMSEA < .06, SRMR < .08
- Small changes (e.g.,  $\Delta CFI \geq -0.01$ ;  $\Delta RMSEA \geq .015$ ;  $\Delta SRMR \geq .0030$ ) support invariance (Chen, 2007)

## PRELIMINARY RESULTS

### MI Testing (Table 2)

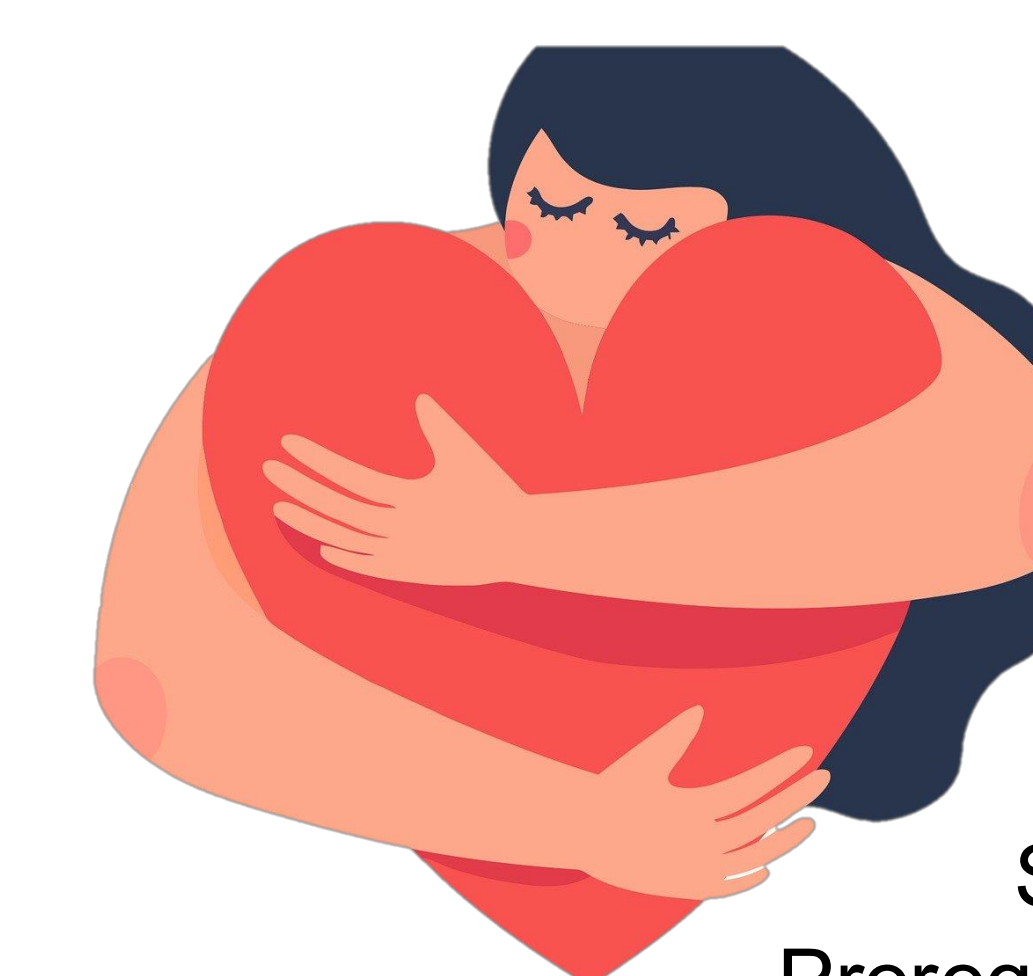
- Relationship history & political orientation: Strict invariance
- Age: Weak invariance
- Gender: Configural invariance for commitment and weak invariance for its predictors

### Group Differences (Figure 1, Table 3)

- Conservatives ( $M = 4.04$ ,  $SD = 1.58$ ) report greater investment into singlehood than liberals ( $M = 3.67$ ,  $SD = 1.49$ ; Cohen's  $d = 0.24$ ).
- Political orientation was positively associated with investment ( $r = 0.13$ ,  $p \leq .001$ ,  $t = 3.60$ )
- No significant differences across relationship history & political orientation for commitment, satisfaction, and alternatives (Cohen's  $d$ s  $\geq 0.07$ )

## DISCUSSION

- We pre-registered collecting an additional sample to ensure power and all samples will be combined for final analyses.
- The findings suggest preliminary evidence of measurement invariance for IMS-S for relationship history and political orientation, and lack evidence for age and gender.
- Non-invariance across gender and age may reflect differences in how individuals interpret or prioritize singlehood, shaped by social norms, life stage, or relational expectations.
- Future research should use conduct differential item functioning analyses and incorporate qualitative methods to explore how individuals interpret items and understand the sources of non-invariance



Scan for  
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